

Persona-L has Entered the Chat: Leveraging LLMs and Ability-based Framework for Personas of People with Complex Needs

Problem

Data gathering:

- Persona creation traditionally relies on data from qualitative methods like interviews, observations and surveys
- Methods to get data for personas are difficult because of practicality and ethical reasons + not the most trustworthy sources sometimes

Persona generation:

- LLMs can generalize complex groups of people with specific needs because their data is not represented in LLM training data

Question: How to leverage LLMs and an ability based framework to build more accurate/encompassing personas of people with complex needs?

Research Gap: LLM generated personas lacked depth and were generic built off templates. Emphasizes real time interaction and persona responses grounded in data for people with Down syndrome in a contextually richer way

What they do

- Ability first framework which looks at what the persona is able to do
 - What blockers make the persona have a more difficult time achieving a task
 - What is helpful
 - Abilities related to employment, education, family
- Collecting data about the abilities and strengths of people with complex needs (e.g. allowing users to select options for the abilities to explore an interface)
- Users able to engage in a conversation with the persona they developed
- RAG prompting - uses a curated knowledge base and allows the LLM to retrieve relevant data from the grounded dataset in real time before generating a response (used RALM model)

Build Sensitive Group Personas

Fill in the details to create your personalized persona

Age

10-20

Occupation

Baker

Medical Condition

Down Syndrome

Theme

Employment

Design Goals

CREATE PERSONA

PERSONA PROFILE

Name: Doreen
Age: 19
Gender: Female
Occupation: Baker
Medical Condition: Down Syndrome
Theme: employment
Doreen is a vibrant 19-year-old from Maplewood, BC, Canada, who has Down syndrome and a deep passion for baking. With dreams of opening her own bakery, she delights in creating sweet treats and sharing her love for the culinary arts. Doreen believes that everyone can pursue their career aspirations, regardless of challenges they may face. Eager to connect with others, she is enthusiastic about discussing employment and sharing her personal experiences, all while spreading positivity and inspiration. 🍰💪

SELECT

Explore Key Abilities

Select and Learn About Key Abilities of Your Persona Based on the Theme

EMPLOYMENT

Memory Skills

Hi! I am Doreen. I have exceptional memory skills that greatly aid me in my work. Despite the challenges posed by Down syndrome, I can effectively retain and recall information, enabling me to perform tasks with accuracy and efficiency. This proficiency enhances my ability to handle complex tasks and contribute effectively to my team.

Explore Ability Drivers and Blockers

+ SELECT

Teamwork

Hi! I am Doreen. I have a strong spirit of teamwork. I collaborate closely with my team members to effectively accomplish my tasks. My ability to communicate and cooperate with others makes me a valuable team player.

Explore Ability Drivers and Blockers

+ SELECT

Willingness to Learn

Hi! I am Doreen. I demonstrate a strong willingness to learn. In my job, I embrace new opportunities for growth and development, continually enhancing my skills and knowledge. This eagerness to learn makes me a valuable asset to my team and the organization.

Explore Ability Drivers and Blockers

+ SELECT

BACK

NEXT

+ New Chat

Previous 7 Days

Chat with Doreen 1

Previous 30 Days

Chat with Doreen about employment

Stereotypes

employment and share my experiences. I'm excited to connect with you! What would you like to know? 🍰💪

How do you manage to remember the detailed steps for different tasks during the week?

I manage to remember the detailed steps for different tasks at work by using picture-based task schedules and checklists. These visual aids help me stay organized and focused, especially during busy times in the bakery. I also benefit from one-on-one training when I'm given new responsibilities, which allows me to practice until I feel comfortable. My supervisors give me instructions one at a time with visual demonstrations, which makes it easier for me to understand what I need to do. Regular feedback from my job coach and supervisor helps me learn and adjust my techniques too. 🍰

Documentation

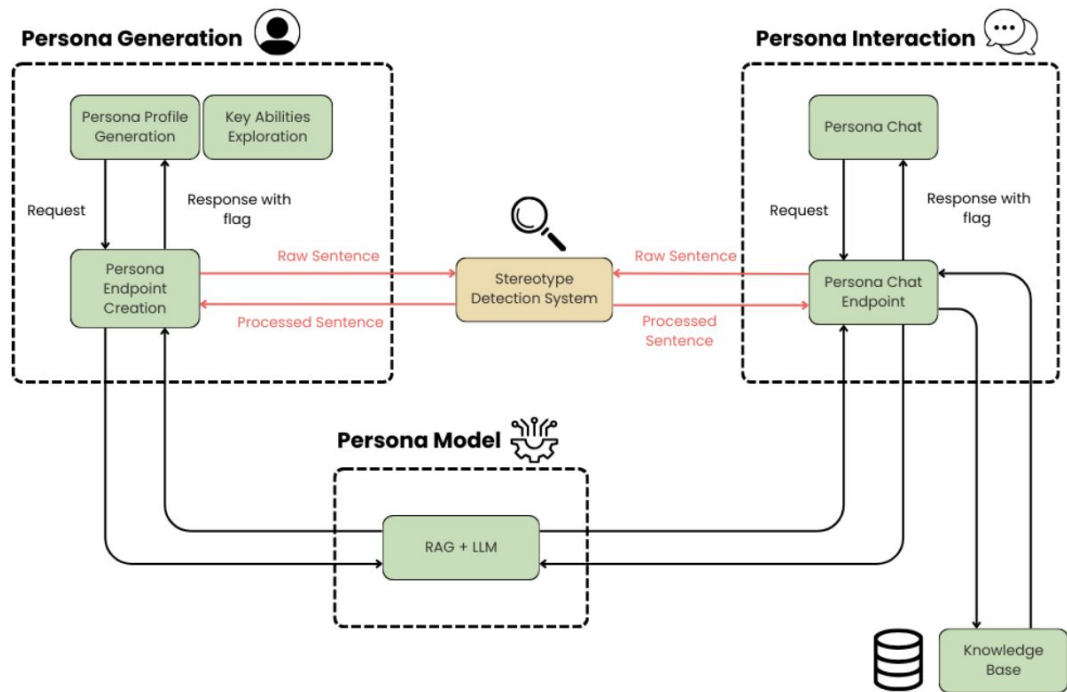
Name: Doreen
Age: 19
Gender: Female
Occupation: Baker
Diagnosis: Down Syndrome

Abilities:

Memory Skills

Marked Content

Summarize your conversation



Evaluation

- Generate believable personas but has bias and hallucinations (makes up fake things)
- persona L’s ability and blockers helped maintain continuous understand of persona context
- Focused too much on positive traits though
- Promotes empathy building
- Not fully balanced humanly
- People feel like their talking to a version of themselves instead of another persona
- Data transparency good but need references for data to build trust

Future Scope (if any)

- Iterate of Persona L to address challenges related to stereotype mitigation and bias so that the persona stays inclusive
- Increase diversity
- Need to regularly update the dataset

Table 1:

Participant ID	Age	Gender	Country	Occupation	Years of Experience
P1	25-34	Female	Canada	Product Designer	3-5
P2	35-44	Male	USA	UX Designer	1-3
P3	25-34	Female	USA	UX Designer/UX Researcher	1-3
P4	25-34	Male	Canada	HCI Lecturer	3-5
P5	25-34	Male	USA	UX Designer	>5
P6	35-44	Female	USA	Senior Product Manager	>5

Does ChatGPT Generate Accessible Code? Investigating Accessibility Challenges in LLM-Generated Source Code

Problem

- LLMS are more often used to generate web code but their ability to produce accessible code is not very explored
- LLMs trained on existing web data which often contains lots of WCAG violations

Research Gap: Studies specifically if LLM generated websites can be accessible/how accessible they are according to WCAG guidelines

- **What they do**
 - 1. Developers prompt GPT to write website code. Websites systematically assessed for compliance with Achecker and WAVE
 - 2. Developers iterate and prompt to fix those issues. Then put back into Achecker and WAVE
 - See if GPT fixes accessibility issues in other code bases from 3rd party websites (88 open source git projects)
 - 84% of the GPT generated websites contained accessibility issues mostly related under Perceivable WCAG principle
 - Contrast, resize, link purpose, and info and relationships are some big ones
 - Seems to have better performance at ensuring website code structure for ATS
- **Evaluation**
 - Accessibility cannot be presumed when asking for GPT generated sites
 - GPT generates websites with wide range of accessibility errors suggesting lack of proper association with expected requirements such as accessibility
 - Need to train more on inclusive designs
- **Future Scope (if any)**
 - Accessible issues in mobile code. Also look into open source LLMS to understand difference in code generation capability
 - analyze patterns in participant prompts used during study to see how developer interactions impact LLM web development context